Week 4 — In-Class Exercise

Features by hand · name the segments · pick the tool

Applied Data Science and AI in Pharma Commercial · Columbia University · Fall 2026

Groups of 3–4 · 20 minutes + 5-minute debrief · Monday, October 5, 2026

Bring a calculator or a notebook cell. Answers on this sheet; one sheet per group, names on the back.

Part A · Features by hand (7 minutes)

Six GLUCOREN prescribers, 2025 totals by half-year. Compute the five features, then rank the six by market volume (a six-HCP 'decile'), then assign each a behavior in one phrase. Where do the two orderings disagree, and why?

HCP	Market TRx H1 / H2	GLUCOREN TRx H1 / H2	NBRx (yr)	Switch-in (yr)	Emails sent / opened	Calls (yr)	% Medicare	Reject rate
Dr. Alvarez (PCP)	520 / 540	190 / 215	62	20	36 / 16	26	28%	5%
Dr. Brennan (Endo)	610 / 590	35 / 38	12	1	36 / 4	20	34%	7%
Dr. Chen (NP)	240 / 260	22 / 52	30	17	36 / 14	15	26%	6%
Dr. Dube (PCP)	400 / 380	52 / 44	21	5	36 / 10	22	63%	32%
Dr. Evans (PCP)	70 / 80	7 / 9	5	1	24 / 5	4	40%	9%
Dr. Farrow (Cardio)	450 / 470	60 / 61	10	1	36 / 5	18	36%	8%

Formulas. Share = brand ÷ market (year). Trend = share(H2) − share(H1), in points. NBRx share = NBRx ÷ brand TRx. Switch-in rate = switch-in ÷ NBRx. Email open rate = opened ÷ sent.

HCP	Share	Trend (pts)	NBRx share	Switch-in rate	Open rate	Volume rank (1 = biggest)	Behavior (one phrase)
Dr. Alvarez (PCP)							
Dr. Brennan (Endo)							
Dr. Chen (NP)							
Dr. Dube (PCP)							
Dr. Evans (PCP)							
Dr. Farrow (Cardio)							

A1. Which HCP would a decile call plan over-serve most? Which would it under-serve?

_____

A2. One of the six has a feature that is not a behavior of the HCP at all. Which, and whose behavior is it?

_____

Part B · Name and activate (8 minutes)

The segment card table from tonight's notebook, with the clusters shuffled and unnamed. For each segment: a name that describes a behavior (never a letter or a number), one play (channel, frequency, message or program) and one exit trigger (the observable event that moves an HCP out).

Segment	HCPs	% of brand TRx	Share	Trend (pts / half-year)	NBRx share	Switch-in	Calls / HCP-yr	Call resp.	Email open	Web visits	Medicare	Reject	Churn-risk share
A	918	16%	16%	+7.7	48%	54%	16.8	0.08	38%	49	28%	7%	0%
B	1,309	15%	6%	+0.6	17%	5%	19.3	0.01	14%	6	35%	7%	3%
C	547	45%	38%	+2.8	32%	36%	26.7	0.17	42%	36	29%	6%	1%
D	2,412	8%	11%	+0.3	34%	13%	4.9	0.00	20%	7	39%	9%	19%
E	814	16%	12%	−0.6	40%	25%	21.7	0.03	30%	21	60%	31%	86%

Segment	Name (a behavior)	Play	Exit trigger
A			
B			
C			
D			
E			

B1. Which two segments could you merge with the least loss? What would you lose?

_____

B2. Segment E has an 86% churn-risk share. Is churn here a messaging problem? Who owns the play?

_____

Part C · Pick the tool (5 minutes)

For each scenario: decile · behavioral clustering · needs-based · supervised score · HDBSCAN · 'not a segmentation'. One sentence of justification.

C1. A rare-disease brand has 380 treating physicians in the US. The brand lead asks for 'a segmentation'.

Tool: _____ Why:

_____

C2. A primary-care brand wants to decide which HCPs get a rep visit next month and which get an email.

Tool: _____ Why:

_____

C3. The agency needs to write three different creative campaigns and wants to know what each audience believes.

Tool: _____ Why:

_____

C4. The sales-ops team must set call frequency and incentive-comp targets for 2027 by January.

Tool: _____ Why:

_____

C5. A launch brand, 9 months in, wants to know which of its 5,000 targets are adopting, which are trying-and-stopping, and which have never written.

Tool: _____ Why:

_____

C6. A brand team suspects that 40% of its universe has no distinct behavior at all and wants the segmentation to say so.

Tool: _____ Why:

_____